**Budget Justification**

## 1 Salaries

Three Ph.D. and two M.Sc. students will perform the different tasks of the research program over a period of five years. Each Ph.D. student will receive a grant of $30,000 per year for a period of four years. Each M.Sc. student will receive an annual grant of $29,000 for two years. The total amount dedicated to student grants is $476,000.

Table 1: Salaries and benefits, divided by task and HQP

| Task | HQP | Year 1 | Year 2 | Year 3 | Year 4 | Year 5 |
|---|---|---|---|---|---|---|
| i1 | PhD 1 | $30,000 | $30,000 | $30,000 | $30,000 | |
| 2 | PhD 2 | $30,000 | $30,000 | $30,000 | $30,000 | |
| 3 | PhD 3 | $30,000 | $30,000 | $30,000 | $30,000 | |
| 4 | MSc 1 | | | | $29,000 | $29,000 |
| | MSc 2 | | | | $29,000 | $29,000 |
| Total | | $90,000 | $90,000 | $90,000 | $148,000 | $58,000 |

## 2 Equipment and supplies

An amount of $1000 per student will be dedicated for the purchase of computers.

## 3 Travel

Each of the three Ph.D students will present their research at international conferences twice. Costs are estimated at $4,000 per conference, including travel, conference fees and accommodations. Potential conferences include the biennial conferences of the International Society for Industrial Ecology (ISIE), the Gordon Research Conference (GRC) on Industrial Ecology, the Life Cycle Management Conference (LCM), and the Conference on Life Cycle Assessment of Waste.

## 4 Dissemination

Each M.Sc. student is expected to publish one article, and each Ph.D. student is expected to publish three articles, for a total of 11 articles. Given the uneven level of English skills among HQPs, it is anticipated that approximately one third of the articles submitted in this program will require professional language editing, and $3,000 are planned for this eventuality.

This research program will comply with the Tri-Agency Open Access Policy on Publication through the self-archive services of Polytechnique Montreal. The applicant therefore does not require funds for covering journal open-access publication fees.

## 5 Material, facility and other expenses

No materials, dedicated facility, or other expenses will be specifically required to complete the activities of this research program.

## 6 Overview

As can be observed from table 2, HQP salaries amount to more than 90% of the requested funds.

Table 2: Total expenses over the 5-year period

| | Amount | Share |
|---|---|---|
| Salaries | $476,000 | 93.7% |
| Equipment and databases | $5,000 | 1,0% |
| Travel and conferences | $24,000 | 4,7% |
| Dissemination | $3,000 | 0.5% |
| Total | $508,000 | |